COMP-551DL “Business Intelligence”
Assignment 1
Customer Churn Analysis and Classification

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Introduction and Objective

This project analyses the IBM Telco Customer Churn dataset to identify customer characteristics associated with churn. It compares two binary classification models using a separated portion test set and ROC curves to assess their ability to distinguish customers who churned from those who stayed.

1. Task 1: Data Acquisition and Analysis

1.1 Dataset source and selection

The IBM Telco Customer Churn dataset was obtained from Kaggle[1]. It contains 7,043 customer records, a binary churn outcome, and demographic, service, and billing information. It was selected for its interpretable business features, manageable size, and suitability for comparing binary classifiers. The dataset is IBM sample data representing a fictional telecommunications company[2].

1.2 Dataset size and feature types

The original dataset contains 7,043 rows and 21 columns, with one customer per row. These comprise 19 potential predictors, the customer identifier customerID, and the binary target Churn (Yes or No). After cleaning, 7,032 records remain.

Numerical features include tenure, MonthlyCharges, and TotalCharges, which was converted from text to numeric. Categorical features describe demographics, subscribed services, contracts, and payment methods. These include binary variables such as Partner and SeniorCitizen, and variables with multiple categories such as Contract, InternetService, and PaymentMethod. Although stored as 0 and 1, SeniorCitizen represents a category.

1.3 Data quality and cleaning

Initial missing-value checks reported no missing values, but a whitespace check identified 11 blank entries in TotalCharges. All belonged to customers with zero tenure. Converting the column using pd.to_numeric(errors="coerce") confirmed that only these 11 entries were invalid.

These records were removed rather than assuming their unknown charges were zero. This excluded approximately 0.16% of the original data, leaving 7,032 records with numeric TotalCharges and no remaining missing values in that column. The original CSV was preserved, and cleaning was applied to a separate DataFrame.

No duplicate customer IDs or identical rows were found. Categorical values showed no obvious inconsistent labels; meaningful categories such as “No internet service” and “No phone service” were retained. A limitation of the cleaning decision is the exclusion of the 11 zero-tenure customers.

1.4 Target distribution and class imbalance

The cleaned dataset is imbalanced: 73.42% stayed and 26.58% churned. Predicting "No" for every customer would achieve 73.42% accuracy while detecting no churners, so accuracy alone is insufficient.

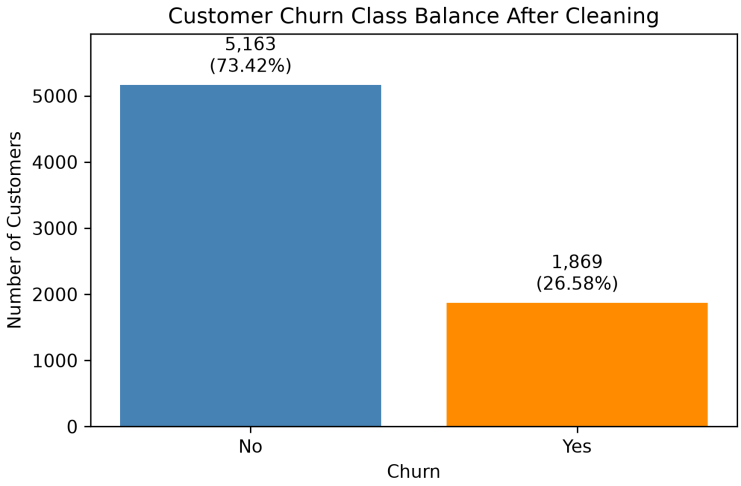


Figure 1. Customer churn distribution after cleaning.

1.5 Exploratory relationships

Month-to-month customers had the highest churn rate (42.71%), compared with 11.28% for one-year and 2.85% for two-year contracts. Customers who churned generally had shorter tenure and higher monthly charges, although both distributions overlap.

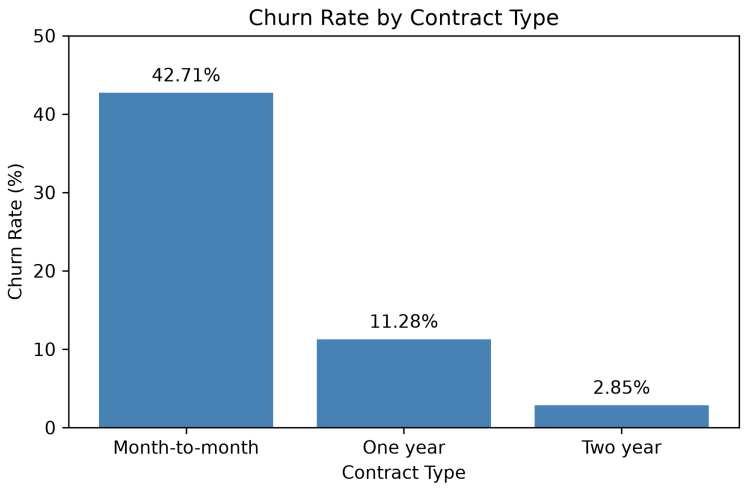


Figure 2. Churn rate by contract type.

Measure	Stayed (No)	Churned (Yes)
Mean tenure (months)	37.65	17.98
Median tenure (months)	38	10
Mean monthly charges	61.31	74.44
Median monthly charges	64.45	79.65

Total charges showed positive Pearson correlations with tenure (0.83) and monthly charges (0.65); tenure and monthly charges had a weaker correlation (0.25).

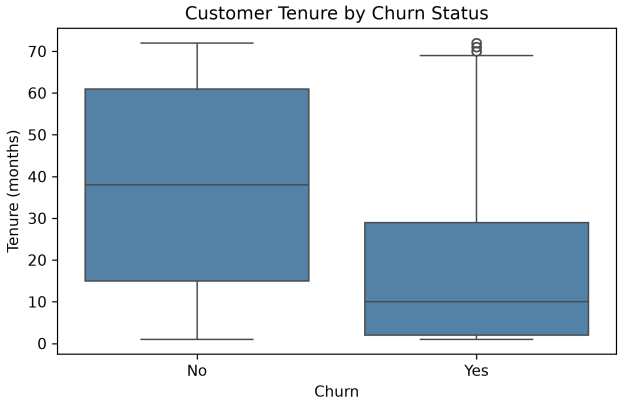


Figure 3. Tenure distribution by churn status.

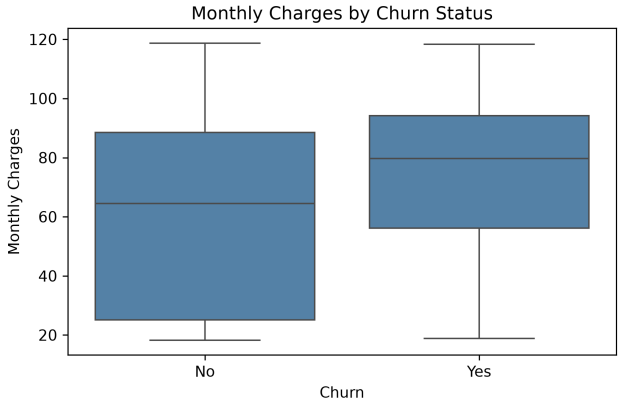


Figure 4. Monthly charges distribution by churn status.

1.6 Sample sufficiency and limitations

The cleaned dataset contains 7,032 customers, including 1,869 churn cases, providing a reasonable starting point for two simple binary classifiers. Sample adequacy also depends on model complexity and the number of features produced after categorical encoding. However, sample size alone does not guarantee reliable performance; this must be assessed using held-out data.

The dataset represents a fictional company and may not reflect other businesses or current customer behaviour. Excluding 11 zero-tenure customers also limits coverage of new customers. Results should be interpreted within this dataset rather than assumed to generalise to real-world churn prediction.

2. Task 2: Fitting Classification Models

2.1 Input features and target

The models use 19 customer features as inputs. customerID was excluded because it is an identifier, and Churn was excluded because it is the target. The target was encoded as 0 for customers who stayed and 1 for customers who churned.

The inputs are gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, and TotalCharges.

2.2 Preprocessing and implementation

The implementation uses Python, pandas, scikit-learn, Matplotlib, and seaborn. ColumnTransformer scales three numerical features, one-hot encodes 15 text features, and keeps SeniorCitizen unchanged. Unseen categories are handled with handle_unknown="ignore".

Each model uses a Pipeline to fit preprocessing and classification on training data only. Test inputs reuse the learned settings. Random Forest uses a separate preprocessor created with clone().

2.3 Logistic Regression

Logistic Regression was used as a simple baseline. It was trained through the preprocessing pipeline with max_iter=1000, allowing sufficient iterations for optimisation.

2.4 Random Forest

Random Forest was selected to capture nonlinear relationships and interactions between features. It used 200 trees, random_state=42, and n_jobs=-1. Both models were trained on the same 5,625 customers.

3. Task 3: Evaluation and Performance Visualisation

3.1 Evaluation framework

An 80/20 stratified split produced 5,625 training and 1,407 test customers, preserving approximately 26.58% churn in each set. The split used random_state=42. Both models used the same split, with preprocessing fitted on training data only. The full dataset had been explored before splitting, but test data was excluded from fitting and tuning.

3.2 Results and ROC curves

Logistic Regression performed better on this test split, achieving ROC-AUC of 0.8359 compared with 0.8128 for Random Forest. Its recall was 57.49%, versus 48.93%, so it identified more actual churners at the default decision threshold.

Both curves lie above the chance diagonal. Logistic Regression generally provides a higher true positive rate at comparable false positive rates. Neither model perfectly separates the classes.

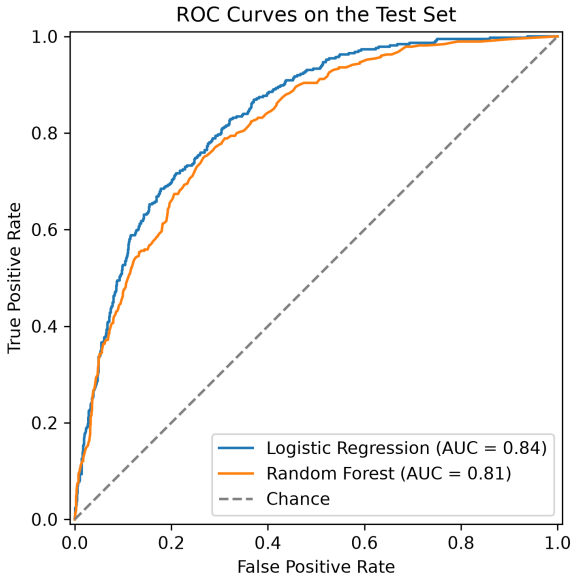


Figure 5. ROC curves for both classifiers on the test set.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8045	0.6495	0.5749	0.6099	0.8359
Random Forest	0.7861	0.6246	0.4893	0.5487	0.8128

predict() produces class predictions, while predict_proba(X_test)[:, 1] provides churn probabilities for ROC-AUC and ROC curves. Classification metrics use the default 0.5 decision boundary.

3.3 Possible improvements

Recent changes in service usage, late payments, complaints, and unresolved support requests could provide additional warning signs of churn. These features must use information available before prediction. Their benefit would need to be tested using validation data from the training set.

3.4 Assumptions and limitations

Customers were treated as independent records suitable for a random stratified split. The analysis assumes that input features would be available before predicting churn, although their timing is not fully documented. Results come from one test split and may vary with a different split. Because the full dataset was explored beforehand, the test set was not completely untouched during analysis.

Conclusion

Churn was associated with month-to-month contracts, shorter tenure, and higher monthly charges. Logistic Regression outperformed Random Forest on this test split, with ROC-AUC of 0.8359 versus 0.8128. However, its recall of 57.49% shows that many churners were still missed. Recent customer behaviour could provide useful additional features, but this would require further testing.

AI Assistance Declaration

ChatGPT and Gemini NotebookLM assisted with project planning, code review, troubleshooting, general models explanations, and output reviews. Assistance focused particularly on model selection, preprocessing concept, training, and evaluation accuracy check.

References

[1] BlastChar. *Telco Customer Churn*. Kaggle.
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

[2] IBM Samples Team. *Telco customer churn (11.1.3+)*. IBM Community, 2019.
https://community.ibm.com/community/user/blogs/steven-macko/2019/07/11/telco-customer-churn-1113